Nicholas Kern

NASA Hubble Fellow

Contact Information

University of Michigan Dept. of Physics
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Employment

NASA Hubble Fellow — September 2023 – present
Department of Physics
University of Michigan, Ann Arbor, MI, USA

Pappalardo Fellow — September 2020 – September 2023
Department of Physics & MIT Kavli Institute for Astrophysics and Space Research
Massachusetts Institute of Technology, Cambridge, MA, USA

Education

Ph.D., Astrophysics, University of California, Berkeley — August 2020
Advisor: Aaron R. Parsons

M.A., Astrophysics, University of California, Berkeley — May 2017

B.S., Physics, Astrophysics, University of Michigan, Ann Arbor — May 2015

Research Interests

cosmological data analysis, star and galaxy formation, large scale structure, machine learning, radio interferometry, astrostatistics

Honors & Awards

Hubble Fellow, NASA	2023
Pappalardo Fellow, MIT, Department of Physics	2020
Mary Elizabeth Uhl Dissertation Prize, UC Berkeley, Department of Astronomy	2020
Teaching Effectiveness Award, UC Berkeley	2017
Outstanding Graduate Student Instructor Award, UC Berkeley	2017
Graduated with Highest Honors and Distinction, University of Michigan	2015
Excellence in Astrophysics Research Award, University of Michigan	2015
International Institute Fellow, University of Michigan	2014
Upper-Level Writing Prize in the Natural Sciences, University of Michigan	2014

Students Advised

- Robert Pascua, McGill PhD Student — Fall 2023 – Spring 2025
  Quantifying signal reconstruction fidelity with radio telescope forward models
- Honggeun Kim, MIT PhD Student — Fall 2021 – Fall 2024
  Mitigating the impact of antenna feed deviations for 21 cm cosmology
- Ntsikelelo Charles, Rhodes U., South Africa, PhD student — Spring 2021 – Fall 2024
  Inteferometric gain calibration in the presence of mutual coupling
  Mitigating diffuse foregrounds for interferometric calibration
- Cynthia De Los Santos, San Bernardino Community College — Summer 2022
  Spherical Harmonic Decomposition of HERA's Primary Beam
- Eleanor Rath, MIT PhD student — Fall 2021 – Fall 2022
  A Bayesian framework for modeling antenna beam perturbations
- Duncan Rocha, Harvey Mudd undergrad (→ U. Chicago grad) — Summer 2017
  Detectability of Alcock Paczynski effects for 21 cm intensity mapping
- Timothy Wilson, UCLA undergrad (→ UCLA grad) — Summer 2016
  An MCMC sampler for semi-numerical Cosmic Dawn simulations

Teaching Experience

- Session Instructor for *Interferometric Calibration and Imaging* — 2018 – 2021
  - Designed and taught a 3-hour lesson for the HERA summer undergraduate bootcamp
- Head Instructor for *Python Programming in Astronomy* at UC Berkeley — Summer 2017
  - Developed a new 6-week intensive undergrad summer course for the Astronomy Dept.
  - Focus on scientific computing, computational physics, & data science
  - Lectured daily, held office hours, wrote and graded midterms, oversaw final projects
- Graduate Instructor for *Introduction to Astrophysics* at UC Berkeley — Fall 2016
  - Led discussion section, developed interactive worksheets, graded homework & exams
  - Awarded department-wide "Outstanding Graduate Instructor" and university-wide "Teaching Effectiveness Award"
- Graduate Instructor for *Stellar Structure & Evolution* at UC Berkeley — Fall 2015
  - Led discussion section, developed interactive worksheets, graded homework & exams
- Undergraduate Instructor for *Introduction to Mechanics* at U. Michigan — Spring 2015
  - Taught undergraduates in breakout coding sessions, held office hours

Service

To the Astrophysics Community:

- Referee, Radio Science — 2020 – present
- Referee, Monthly Notices of the Royal Astronomical Society — 2019 – present
- Referee, Astrophysical Journal — 2018 – present

At the University of Michigan

- LOC for the 2025 Michigan Summer Cosmology School — 2025

At the Massachusetts Institute of Technology

- Primary Project Advisor, HERA Undergraduate Summer Research Bootcamp — 2022
- Lead Coordinator, HERA Undergraduate Summer Research Bootcamp — 2021
- Instructor & Mentor, HERA Undergraduate Summer Research Bootcamp — 2020 – 2021

At the University of California, Berkeley

- Graduate Representative, UC Berkeley Faculty Search Committee — 2020
- Instructor & Mentor, HERA Undergraduate Summer Research Bootcamp — 2017 – 2019
- Organizer, Astronomy Career Development Seminar — 2016 – 2017
- Organizer, Graduate Student Colloquium Speaker Seminar — 2015 – 2016

Grants and Compute Allocations

Principal Investigator, *Scaling-Up Differentiable Pipelines for Radio Surveys* — 2025
ACCESS, NCSA Delta Cluster

Principal Investigator, *Bayesian Frameworks for New 21 cm Telescopes* — 2022
XSEDE, PSC Bridges2 Cluster

Principal Investigator, *ML Tools for 21 cm Constraints on Fundamental Physics* — 2022
NERSC, Perlmutter Cluster

Publications Led, Collaboration-Equivalent, or Primary Student* Mentor

13. **Kern, N.** 2025, *A Differentiable, End-to-End Forward Model for 21 cm Cosmology: Estimating the Foreground, Instrument, and Signal Joint Posterior*, MNRAS 541 687K

12. Pascua, R.*, Martinot, Z., Liu, A., Aguirre, J., **Kern, N.**, et al. 2024, *A Generalized Method for Characterizing 21-cm Power Spectrum Signal Loss from Temporal Filtering of Drift-scanning Visibilities*, ApJ 985 127P

11. Charles, N.*, **Kern, N.**, Pascua, R., et al. 2024, *Mitigating calibration errors from mutual coupling with time-domain filtering*, MNRAS 534 3349C

10. Kim, H.*, **Kern, N.**, Hewitt, J., et al. 2023, *The Impact of Beam Variations on Power Spectrum Estimation for 21 cm Cosmology. II. Mitigation of Foreground Systematics for HERA*, ApJ 953 136

9. Charles, N.*, **Kern, N.**, Bernardi, G. et al. 2023, *On the use of temporal filtering for mitigating galactic synchrotron calibration bias in 21 cm reionization observations*, MNRAS 522 1009C

8. Barry, N., Bernardi, G., Greig, B., **Kern, N.** (lead author) and Mertens, F. 2022, *SKA-Low Intensity Mapping Pathfinder Updates: Deeper 21 cm Power Spectrum Limits from Improved Analysis Frameworks*, JATIS 8(1) 011007

7. HERA Collaboration 2022, including **Kern, N.** (lead author), *First Results from HERA Phase I: Upper Limits on the Epoch of Reionization 21 cm Power Spectrum*, ApJ 925 221A

6. **Kern, N.** & Liu, A. 2021, *Gaussian Process Foreground Subtraction and Power Spectrum Estimation for 21 cm Cosmology*, MNRAS 501 1463K

5. **Kern, N.**, Dillon, J. S., Parsons, A. R., Carilli, C., Bernardi, G. et al. 2020, *Absolute Calibration Strategies for the Hydrogen Epoch of Reionization Array and Their Impact on the 21 cm Power Spectrum*, ApJ 890 122

4. **Kern, N.**, Parsons, A. R., Dillon, J. S., Lanman, A. E., et al. 2020, *Mitigating Internal Instrument Coupling for 21cm Cosmology. II. A Method Demonstration with the Hydrogen Epoch of Reionization Array*, ApJ 888 70

3. **Kern, N.**, Parsons, A. R., Dillon, J. S., Lanman, A. E., Fagnoni, N. and de Lera Acedo, E. 2019, *Mitigating Internal Instrument Coupling for 21cm Cosmology. I. Temporal and Spectral Modeling in Simulations*, ApJ 884 105

2. **Kern, N.**, Liu, A., Parsons, A. R., Mesinger, A., & Greig, B. 2017, *Emulating Simulations of Cosmic Dawn for 21 cm Power Spectrum Constraints on Cosmology, Reionization and X-ray Heating*, ApJ 848 23

1. **Kern, N. S.**, Keown, J. A., Tobin, J. J., Mead, A., & Gutermuth, R. 2016, *Radio Properties of Young Stellar Objects in the Serpens South Infrared Dark Cloud*, AJ 151 42

Other Publications as a Contributing Author

27. Chen, K., Wilensky, M., Liu, A., ..., **Kern., N.** et al. 2024, *Impacts and Statistical Mitigation of Missing Data on the 21cm Power Spectrum: A Case Study with the Hydrogen Epoch of Reionization Array*, ApJ 979 191C

26. Rath, E., Pascua, R., Josaitis, A., ..., **Kern., N.** et al. 2024, *Investigating Mutual Coupling in the Hydrogen Epoch of Reionization Array and Mitigating its Effects on the 21-cm Power Spectrum*, Submitted to MNRAS

25. Murphy, G., Bull, P., Santos, M., ..., **Kern., N.** et al. 2023, *Bayesian estimation of cross-coupling and reflection systematics in 21cm array visibility data*, MNRAS 534 2653M

24. Berkhout, L., Jacobs., D., ..., **Kern, N.**, et al. 2024, *Hydrogen Epoch of Reionization Array (HERA) Phase II Deployment and Commissioning*, PASP 136d 5002B

23. Keller, P., Nikolic, B., Thyagarajan, N., ..., **Kern, N.**, et al. 2023, *Search for the Epoch of Reionisation with HERA: Upper Limits on the Closure Phase Delay Power Spectrum*, MNRAS 524 583K

22. HERA Collaboration 2023, including **Kern, N.**, *Improved Constraints on the 21 cm EoR Power Spectrum and the X-Ray Heating of the IGM with HERA Phase I Observations*, ApJ 945 124

21. Pagano, M., Liu, J., Liu, A., **Kern, N.** et al. 2022, *Characterization Of Inpaint Residuals In Interferometric Measurements of the Epoch Of Reionization*, MNRAS 520 5552P

20. Kim, H., Nhan, B., Hewitt, J., **Kern, N.** et al. 2022, *The Impact of Beam Variations on Power Spectrum Estimation for 21-cm Cosmology I: Simulations of Foreground Contamination for HERA*, ApJ 941 207

19. Xu, Z., Hewitt, J., ..., **Kern, N.** et al. 2022, *Direct Optimal Mapping for 21cm Cosmology: A Demonstration with the Hydrogen Epoch of Reionization Array*, ApJ 938 128

18. HERA Collaboration 2022, including **Kern, N.**, *HERA Phase I Limits on the Cosmic 21-cm Signal: Constraints on Astrophysics and Cosmology During the Epoch of Reionization*, ApJ 924 51A

17. Aguirre, J., Murray, S., ..., **Kern, N.**, et al. 2021, *Validation of the HERA Phase I Epoch of Reionization 21 cm Power Spectrum Software Pipeline*, ApJ 924 85A

16. LaPlante, P., Williams, P. K. G., ..., **Kern, N.**, et al. 2021, *A Real Time Processing System for Big Data in Astronomy: Applications to HERA*, A&C 3600489L

15. Tan, J., Liu, A., **Kern, N.**, et al. 2021, *Methods of Error Estimation for Delay Power Spectra in 21cm Cosmology*, ApJS 255 26T

14. Ewall-Wice, A., **Kern, N.**, Dillon, J. S., et al. 2021, *DAYENU: A Simple Filter of Smooth Foregrounds for Intensity Mapping Power Spectra*, MNRAS 500 5195E

13. Nunhokee, C. D., Parsons, A. R., **Kern, N.**, et al. 2020, *Measuring HERA's primary beam in-situ: methodology and first results*, ApJ 897 5N

12. Thyagarajan, N., Carilli, C., Nikolic, B., ..., **Kern, N.**, et al. 2020, *Detection of Cosmic Structures using the Bispectrum Phase. II. First Results from Application to Cosmic Reionization Using the Hydrogen Epoch of Reionization Array*, Phys. Rev. D 102, 022002

11. Dillon, J. S., Lee, M., Ali, Z. S., ..., **Kern, N.**, et al. 2020, *Redundant-Baseline Calibration of the Hydrogen Epoch of Reionization Array*, MNRAS 499 5840D

10. Ghosh, A., Mertens, F., Bernardi, G., ..., **Kern, N.**, et al. 2020, *Foreground modelling via Gaussian process regression: an application to HERA data*, MNRAS 495 2813G

9. Carilli, C., Thyagarajan, N., Kent, J., ..., **Kern, N.**, et al. 2020, *Imaging and Modeling Data from the Hydrogen Epoch of Reionization Array*, ApJS 247 67

8. Lanman, A. E., Pober, J. C., **Kern, N.**, et al. 2020, *Quantifying EoR delay spectrum contamination from diffuse radio emission*, MNRAS 494 3712L

7. Monsalve, R. A., Greig, B., Bowman, J. D., ..., **Kern, N.**, et al. 2018, *Results from EDGES High-Band: II. Constraints on Parameters of Early Galaxies*, ApJ 863 11

6. Kohn, S. A., Aguirre, J. E., La Plante, P., ..., **Kern, N.**, et al. 2018, *The HERA-19 Commissioning Array: Direction Dependent Effects*, ApJ 882 58K

5. Dillon, J. S., Kohn, S. A., Parsons, A. R., ..., **Kern, N.**, et al. 2017, *Polarized redundant-baseline calibration for 21 cm cosmology without adding spectral structure*, MNRAS 477 5670

4. Miller, C. J., Stark, A., Gifford D., & **Kern, N.** 2016, *Inferring Gravitational Potentials from Mass Densities in Cluster-Sized Halos*, ApJ 822 41

3. Stark, A., Miller, C. J., **Kern, N.**, Gifford, D., et al. 2016, *Probing Theories of Gravity with Phase Space-Inferred Potentials of Galaxy Clusters*, Phys. Rev. D 93, 084036

2. Gifford, D., **Kern, N.**, & Miller, C. 2016, *Stacking Caustic Masses from Galaxy Clusters*, ApJ 834 204

1. Gifford, D., Miller, C. J., & **Kern, N.** 2013, *A Systematic Analysis of Caustic Methods for Galaxy Cluster Masses*, ApJ 773 116

Collaboration Publications

6. Kittiwisit, P., Murray, S., Garsden, H., ..., **Kern, N.** et al. 2023, *matvis: A matrix-based visibility simulator for fast forward modelling of many-element 21 cm arrays*, RASTI 4 1K
5. Gorce, A., Ganjam, S., Liu, A., ..., **Kern, N.**, et al. 2022, *Impact of instrument and data characteristics in the interferometric reconstruction of the 21?cm power spectrum*, MNRAS 520 375G
4. Storer, D., Dillon, J., Jacobs, D., ..., **Kern, N.**, et al. 2021, *Automated Detection of Antenna Malfunctions in Large-N Interferometers: A Case Study with the Hydrogen Epoch of Reionization Array*, RaSc 5707376S
3. Gehlot, B., Jacobs, D., ..., **Kern, N.**, et al. 2021, *Effects of model incompleteness on the drift-scan calibration of radio telescopes*, MNRAS 506 4578G
2. Fagnoni, N., de Lera Acedo, E., ..., **Kern, N.**, et al. 2021, *Understanding the HERA Phase I receiver system with simulations and its impact on the detectability of the EoR delay power spectrum*, MNRAS 500 1232F
1. Kerrigan, J., La Plante, P., ..., **Kern, N.**, et al. 2019, *Optimizing sparse RFI prediction using deep learning*, MNRAS 488 2605

Selected Talks

NASA Hubble Symposium — October 2025
Baltimore, MD

U. Toledo Astronomy Colloquium — September 2025
Toledo, OH

Michigan HEP-Astro Seminar — September 2025
Ann Arbor, MI

UIUC Astrophysics & Cosmology Seminar — September 2025
Urbana-Champaign, IL

Wisconsin Cosmology Seminar — May 2025
Madison, WI

Line Intensity Mapping Conference — December 2024
Heidelberg, Germany

NASA Hubble Symposium — September 2024
Los Angeles, CA

Johns Hopkins Theoretical Cosmology Seminar — December, 2023
Baltimore, MD

NASA Hubble Symposium — September, 2023
Cambridge, MA

Wisconsin 21 cm Workshop — August, 2022
Madison, Wisconsin

3rd URSI Atlantic Radio Science Meeting — June 2022
Gran Canaria, Spain

ASTRON/JIVE Colloquium — March 2022
ASTRON, Dwingeloo, Netherlands

Royal Astronomical Society Specialist Discussion — December 2021
Virtual

Science at Low Frequencies VIII — December 2021
Virtual

Astrophysics Seminar — November 2021
Johns Hopkins University, Baltimore, MA

INAF Joint Astrophysical Colloquium INAF, Bologna, Italy	November 2021
Pappalardo Research Symposium MIT, Cambridge, MA, USA	May 2021
A Precursor View of the SKA Sky Virtual	March 2021
Observing the First Billion Years IIT Indore, Indore, India	January 2020
Observational Cosmology Seminar California Institute of Technology, Pasadena, CA	December 2019
Center for Astrophysics SMA Seminar Center for Astrophysics, Cambridge, MA, USA	November 2019
MIT Kavli Institute Brown Bag Lunch Talks MIT, Cambridge, MA, USA	November 2019
BCCP Cosmology Workshop University of California, Berkeley, CA, USA	January 2018
JILA Astrophysics Seminar University of Colorado, Boulder, CO, USA	October 2017
NASA Machine Learning Workshop Mountain View, CA, USA	August 2017